

FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING

Department of Computer Engineering

Academic Year 2025-2026

Rubrics for Lab Experiments

Class: B.E. Computer
Engineering

Subject Name: SMA Lab

Semster: VIII

Subject Code: CSDL8023

Practical No.	4
Title:	Exploratory Data Analysis and visualization of Social Media Data for business.
Date of Performance:	28-11-2025
Roll No.	9913
Student	Mark Lopes

Evaluation

Performance Indicator	Below Average	Average	Good	Excellent	Marks
On Time Submission (2)	Not submitted(0)	Submitted after deadline (1)	Early or on time submission (2)	—	
Test cases and output (4)	Incorrect output (1)	The expected output is verified only a for few test cases (2)	The expected output is Verified for all test cases but is not presentable (3)	Expected output is obtained for all test cases. Presentable and easy to follow (4)	
Coding efficiency (2)	The code is not structured at all (0)	The code is structured but not efficient (1)	The code is Structured and efficient. (2)	-	
Knowledge(2)	Basic concepts not clear (0)	Understood the basic concepts (1)	Could explain the concept with suitable example (1.5)	Could relate the theory with real world application(2)	
Total					

Signature of the Teacher:

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import plotly.express as px
import warnings
warnings.filterwarnings('ignore')

# -----
# 1. MANUALLY CREATED SAMPLE DATASET
# -----

data = {
    "Tweet_ID": range(1, 31),
    "Entity": [
        "PlayStation", "Xbox", "Nintendo", "Steam", "Valorant", "Fortnite",
        "PlayStation", "Xbox", "Nintendo", "Steam", "Valorant", "Fortnite",
        "PlayStation", "Xbox", "Nintendo", "Steam", "Valorant", "Fortnite",
        "PlayStation", "Xbox", "Nintendo", "Steam", "Valorant", "Fortnite",
        "PlayStation", "Xbox", "Nintendo", "Steam", "Valorant", "Fortnite"
    ],
    "Sentiment": [
        "Positive", "Negative", "Neutral", "Positive", "Negative", "Positive",
        "Neutral", "Positive", "Positive", "Neutral", "Negative", "Neutral",
        "Negative", "Positive", "Neutral", "Positive", "Positive", "Negative",
        "Neutral", "Neutral", "Positive", "Negative", "Positive", "Neutral",
        "Positive", "Negative", "Positive", "Neutral", "Negative", "Positive"
    ],
    "Tweet_Content": [
        "Great console!", "Bad experience", "Okay product", "Loved it", "Laggy game",
        "Fine", "Super smooth", "Very fun", "Decent", "Unstable servers", "Good",
        "Not good", "Excellent", "No issues", "Awesome", "Fun gameplay", "Not worki",
        "Alright", "Meh", "Very good", "Poor performance", "Enjoyable", "Good updat",
        "Cool", "Worst service", "Loved it a lot", "Fine game", "Buggy", "Great imp"
    ]
}

df = pd.DataFrame(data)
print(df.head())
```

	Tweet_ID	Entity	Sentiment	Tweet_Content
0	1	PlayStation	Positive	Great console!
1	2	Xbox	Negative	Bad experience
2	3	Nintendo	Neutral	Okay product
3	4	Steam	Positive	Loved it
4	5	Valorant	Negative	Laggy game

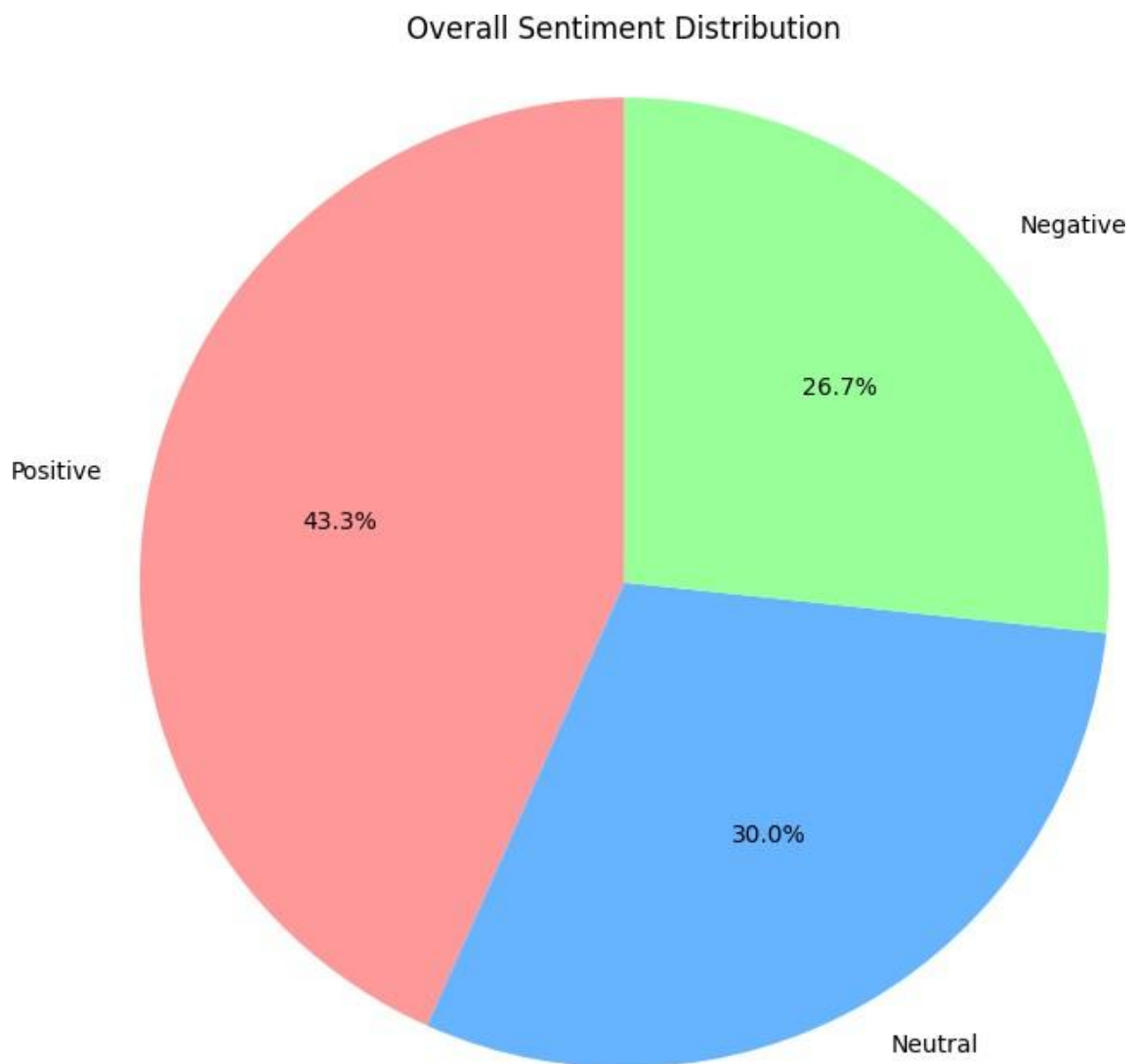
```
In [3]: # -----
# Select Top 6 Entities
# -----

top_entities = df['Entity'].value_counts().head(6).index
df_filtered = df[df['Entity'].isin(top_entities)]

print("\nTop Entities:", top_entities.tolist())
```

Top Entities: ['PlayStation', 'Xbox', 'Nintendo', 'Steam', 'Valorant', 'Fortnite']

```
In [4]: # -----  
# 3.1 PIE CHART - SENTIMENT DISTRIBUTION  
# -----  
  
plt.figure(figsize=(8,8))  
sentiment_counts = df['Sentiment'].value_counts()  
  
plt.pie(sentiment_counts.values,  
        labels=sentiment_counts.index,  
        autopct='%1.1f%%',  
        colors=['#ff9999', '#66b3ff', '#99ff99', '#ffcc99'],  
        startangle=90)  
  
plt.title('Overall Sentiment Distribution')  
plt.axis('equal')  
plt.show()
```



```

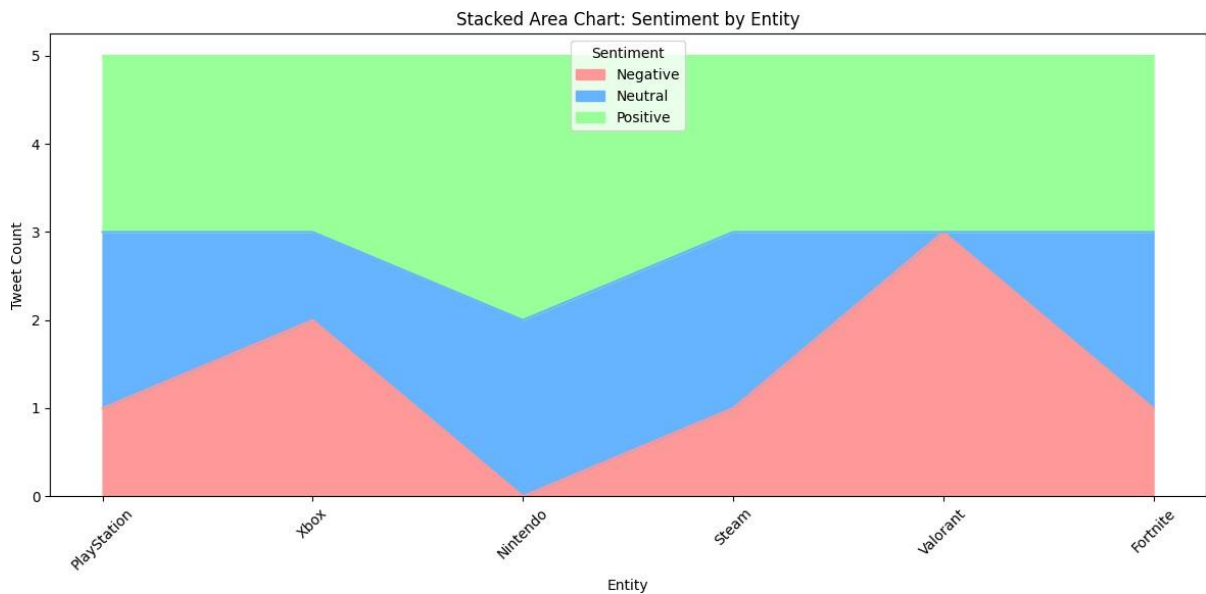
In [5]: # -----
# 3.2 STACKED AREA CHART - BY ENTITY
# -----

pivot_stacked = pd.crosstab(df_filtered['Entity'], df_filtered['Sentiment'])
pivot_stacked = pivot_stacked.reindex(top_entities)

pivot_stacked.plot(kind='area',
                    stacked=True,
                    figsize=(12,6),
                    color=['#ff9999', '#66b3ff', '#99ff99', '#ffcc99'])

plt.title('Stacked Area Chart: Sentiment by Entity')
plt.xlabel('Entity')
plt.ylabel('Tweet Count')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [6]: # -----
# 3.3 RIBBON (CUMULATIVE AREA) CHART
# -----

df_ribbon = df_filtered.groupby(['Entity', 'Sentiment']).size().unstack(fill_value=0)
df_ribbon = df_ribbon.cumsum()

fig = px.area(df_ribbon.reset_index(),
              x='Entity',
              y=df_ribbon.columns,
              title='Ribbon Chart - Cumulative Sentiment Flow',
              labels={'value': 'Count', 'variable': 'Sentiment'},
              color_discrete_sequence=['#ff9999', '#66b3ff', '#99ff99', '#ffcc99'])

fig.show()

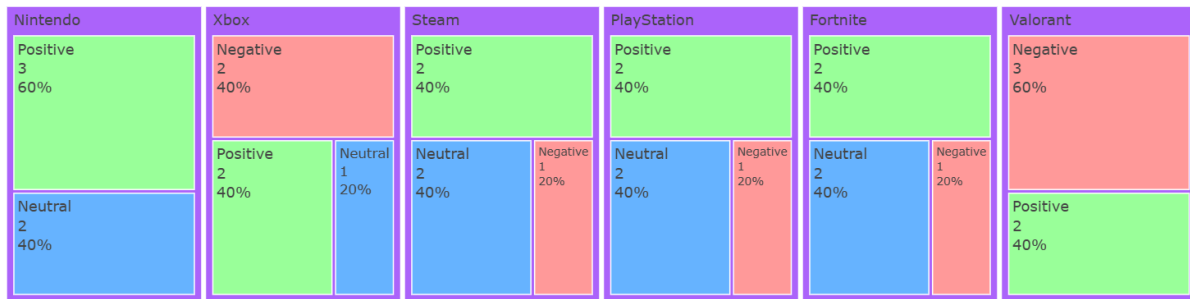
```



```
In [7]: # -----
# 3.4 TREE MAP - ENTITY → SENTIMENT
# -----
fig = px.treemap(df_filtered,
path=['Entity', 'Sentiment'], color='Sentiment', color_discrete_map={
'Positive': '#99ff99', 'Negative': '#ff9999', 'Neutral': '#66b3ff'
}),
title='Tree Map: Sentiment by Entity')

fig.update_traces(textinfo="label+percent parent+value") fig.show()
```

Tree Map: Sentiment by Entity



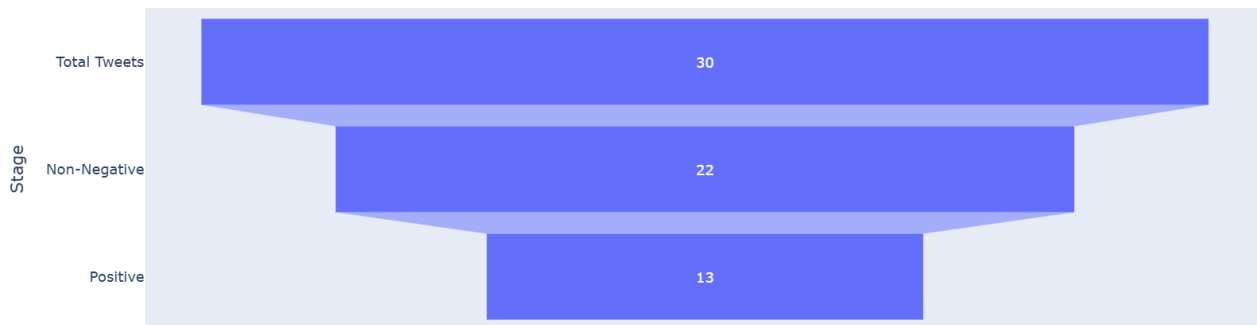
```
In [8]: # -----
# 3.5 FUNNEL CHART - SENTIMENT FILTER
# -----

funnel_data = pd.DataFrame({
    'Stage': ['Total Tweets', 'Non-Negative', 'Positive'],
    'Count': [
        len(df_filtered),
        len(df_filtered[df_filtered['Sentiment'] != 'Negative']),
        len(df_filtered[df_filtered['Sentiment'] == 'Positive'])
    ]
})

fig = px.funnel(funnel_data,
                x='Count',
                y='Stage',
                title='Sentiment Funnel Chart')

fig.show()
```

Sentiment Funnel Chart



```
In [9]: # -----
# 3.6 LINE CHART - SENTIMENT TREND BY ENTITY
# -----

sentiment_by_entity = df_filtered.groupby(['Entity', 'Sentiment']).size().unstack(fill_value=0)
sentiment_by_entity = sentiment_by_entity.loc[top_entities]

plt.figure(figsize=(12,6))
for sentiment in sentiment_by_entity.columns:
    plt.plot(sentiment_by_entity.index,
             sentiment_by_entity[sentiment],
             marker='o',
             label=sentiment)

plt.title('Sentiment Trend Across Entities')
plt.xlabel('Entity')
plt.ylabel('Tweet Count')
plt.xticks(rotation=45)
plt.legend()
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

